

KILLARA HIGH SCHOOL



YEAR 11 – 2010

HSC MATHEMATICS TASK 1

Time allowed: 70 minutes

Total marks: 45

- NOT ALL questions are of equal value.
- Start each question on a new page.
- Show all necessary working.

Question 1: (14 marks)

Marks

a) For what values of k will $3x^2 - kx + 12$ be positive definite 2

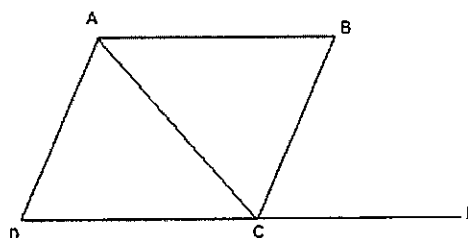
b) Write $8x^2 + 6x + 3$ in the form of $A(2x + 1)^2 + B(2x + 1) + C$ 3

c) In the diagram ABCD is a rhombus where angle $DAC = 54^\circ$ and DC is produced to E.

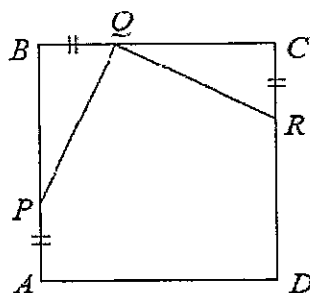
i) What is the value of angle DAB? 1

ii) What is the value of angle BCE? 2

Give reasons for your answers.



d) In the diagram, ABCD is a square. The points P, Q and R lie on AB, BC and CD respectively, such that $AP = BQ = CR$. Prove that:

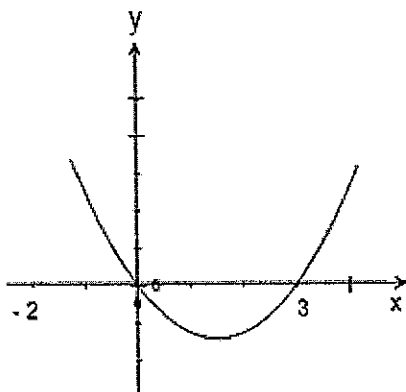


i) triangles PBQ and QCR are congruent. 4

ii) $\angle PQR$ is a right angle. 2

Question 2: (31 marks)**Marks**Differentiate with respect to x (note: marks will be deducted for incorrect notations)

- i) $5x^3 + 6\sqrt{x} - 9$ 2
- ii) $\left(x^2 + \frac{1}{x}\right)(3x^2 - 7x)$ 2
- iii) $\frac{3x^4 - 8}{5 - 2x}$ (leave your answer in the simplest form) 3
- iv) $f(x) = (4x^3 - 7)^5$ 2
- a) Using first principles, differentiate the function $f(x) = x^2 - 3x$ 3
- b) Find the equation of the tangent to the curve $y = 3x^2$ at the point $(-1, 3)$. 2
- c) The diagram shows the graph of the gradient function of the curve $y = f(x)$.
- i) For what values of x is the graph of $f(x)$ increasing? 1
- ii) For what value of x does $f(x)$ have a local maximum? Justify your answer. 2

**Question 2 continue next page**

d) A section of wire fencing 18 metres long is erected into the form of a rectangle

Containing 3 sides only, the fourth side is left open.

If the length of the rectangle is x metres,

i) Show that an expression for the area of the rectangle is $y = 9x - \frac{x^2}{2}$ **2**

ii) Find the maximum possible area. **3**

e) The function $f(x) = 6x^2 - x^3$ and is defined in the domain $-1 \leq x \leq 7$

i) Find the coordinates of any stationary points, and determine their nature. **3**

ii) Determine the coordinates of the point/s of inflexion. **2**

iii) Draw a neat sketch of the function, showing all the above features on your sketch. **3**

iv) In the domain $-1 \leq x \leq 7$, find the greatest value of $f(x) = 6x^2 - x^3$ **1**

Q1 (a) For $3x^2 - kx + 12$ to be positive definite

$$\Delta < 0$$

$$\therefore k^2 - 4 \times 3 \times 12 < 0 \quad \checkmark$$

$$k^2 < 144$$

$$-12 < k < 12 \quad \checkmark$$

(b) $8x^2 + 6x + 3 \equiv A(2x+1)^2 + B(2x+1) + C$

$$\text{L.H.S.} = 4Ax^2 + 4Ax + A + 2Bx + B + C$$

$$= 4Ax^2 + (4A+2B)x + A+B+C \quad \checkmark$$

equate the coeffs. of L.H.S. & R.H.S

$$\Rightarrow 4A = 8$$

$$4A+2B = 6$$

$$A = 2$$

$$2B = 6 - 4 \times 2$$

$$= -2$$

$$B = -1 \quad \checkmark$$

$$C + A + B = 3$$

$$C = 3 - 2 + 1$$

$$= 2$$

$$\therefore 8x^2 + 6x + 3 \equiv 2(2x+1)^2 - (2x+1) + 2 \quad \checkmark$$

(c) i)

$$\angle DAB = 2 \times 54$$

$$= 108^\circ$$

(Diagonals of rhombus bisect the angles) $\checkmark$

ii)

$$\angle DAB = \angle DCB \quad (\text{opp. angles of rhombus are equal})$$

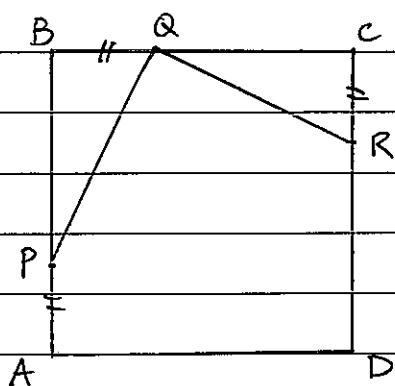
$$= 108^\circ \quad \checkmark$$

$$\angle BCE = 180^\circ - \angle DCB \quad (\text{angles on straight line})$$

$$= 180^\circ - 108^\circ$$

$$\therefore \angle BCE = 72^\circ \quad \checkmark$$

(d)



i)

IN $\triangle PBQ$ & $\triangle QCR$

$$BQ = CR \quad (S) \quad \checkmark$$

$$\angle QBP = \angle QCR$$

$$= 90^\circ \quad (\text{property of a square}) (A) \quad \checkmark$$

$$BC = AB \quad (\text{sides of square})$$

$$BP = AB - PA$$

$$QC = BC - BQ$$

$$\text{but } PA = BQ \quad (\text{given}) \quad \checkmark$$

$$\therefore BP = QC \quad (S)$$

$$\therefore \triangle PBQ \equiv \triangle QCR \quad (SAS) \quad \checkmark$$

ii)

Hence corresponding angles are equal in congruent $\triangle$ s.

$$\Rightarrow \angle BQP = \angle CRQ \quad (1)$$

$$\text{But } \angle CRQ + \angle CQR = 90^\circ \quad (\text{angle sum in a right-angled } \triangle) \quad \checkmark$$

$$\therefore \angle CRQ + \angle BQP = 90^\circ \quad (\text{using } (1)) \quad \checkmark$$

$$\therefore \angle PQR = 180^\circ - \angle CRQ - \angle BQP$$

$$= 90^\circ \quad (\text{angles on a straight line}) \quad \checkmark$$

$$(a) \quad f'(x) = 15x^2 + \frac{3}{\sqrt{x}}$$

$$ii) \quad f'(x) = \left(2x - \frac{1}{x^2}\right)(3x^2 - 7x) + \left(x^2 + \frac{1}{x}\right)(6x - 7)$$

$$iii) \quad f'(x) = \frac{12x^3(5-2x) + 2(3x^4 - 8)}{(5-2x)^2}$$

$$= \frac{60x^3 - 24x^4 + 6x^4 - 16}{(5-2x)^2}$$

$$= \frac{60x^3 - 18x^4 - 16}{(5-2x)^2}$$

iv)

$$f'(x) = 5 \times 12x^2 (4x^3 - 7)^4$$

$$= 60x^2 (4x^3 - 7)^4$$

(b)

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) - x^2 + 3x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3x - 3h - x^2 + 3x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 3)}{h}$$

$$= 2x - 3$$

$$(c) \quad \frac{dy}{dx} = 6x$$

$$\text{at } x = -1, \quad m = 6x - 1$$

$$= -6 \quad \checkmark$$

equation of the tangent:

$$y - 3 = -6(x + 1)$$

$$y = -6x - 3 \quad \checkmark$$

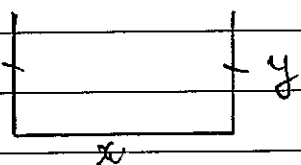
(d)

i) $f(x)$ is increasing for $x < 0$ & $x > 3$ $\checkmark$

ii) $f(x)$ has a max. turning point at $x = 0$ $\checkmark$

because at $x = 0$, $\begin{cases} f'(x) = 0 \text{ and} \\ f''(x) < 0 \end{cases}$ $\checkmark$

(e)



$$2y + x = 18$$

$$\therefore y = \frac{18 - x}{2} = 9 - \frac{x}{2} \quad \checkmark$$

Area of the rectangle is

$$A = x \left(9 - \frac{x}{2} \right)$$

$$= 9x - \frac{x^2}{2} \quad \checkmark$$

$$\text{ii) } \frac{dA}{dx} = 9 - x$$

$$\text{for max. point } \frac{dA}{dx} = 0 \quad \checkmark$$

$$\Rightarrow 9 - x = 0$$
$$x = 9$$

$$\frac{d^2A}{dx^2} = -1 < 0 \quad \Downarrow \quad \checkmark$$

$\therefore$ at $x = 9$ is a maximum

$\therefore$ The max. possible area is

$$A = 9 \times 9 - \frac{9^2}{2}$$
$$= 40.5 \text{ m}^2 \quad \checkmark$$

(e)

$$\text{i) } f'(x) = 12x - 3x^2$$
$$f''(x) = 12 - 6x$$

for stationary points $f'(x) = 0$

$$\Rightarrow 3x(4 - x) = 0$$

$$\begin{cases} x = 0 \\ y = 0 \end{cases}, \begin{cases} x = 4 \\ y = 32 \end{cases} \quad \checkmark$$

$$f''(0) = 12 > 0 \quad \Downarrow$$

$\therefore (0, 0)$ is a minimum point $\checkmark$

$$f''(4) = 12 - 6 \times 4$$
$$= -12 < 0 \quad \Downarrow$$

$\therefore (4, 32)$ is a max. point $\checkmark$

ii) for point of inflexion $f''(x) = 0$

$$\Rightarrow 12 - 6x = 0 \quad \Rightarrow \begin{cases} x = 2 \\ y = 16 \end{cases}$$

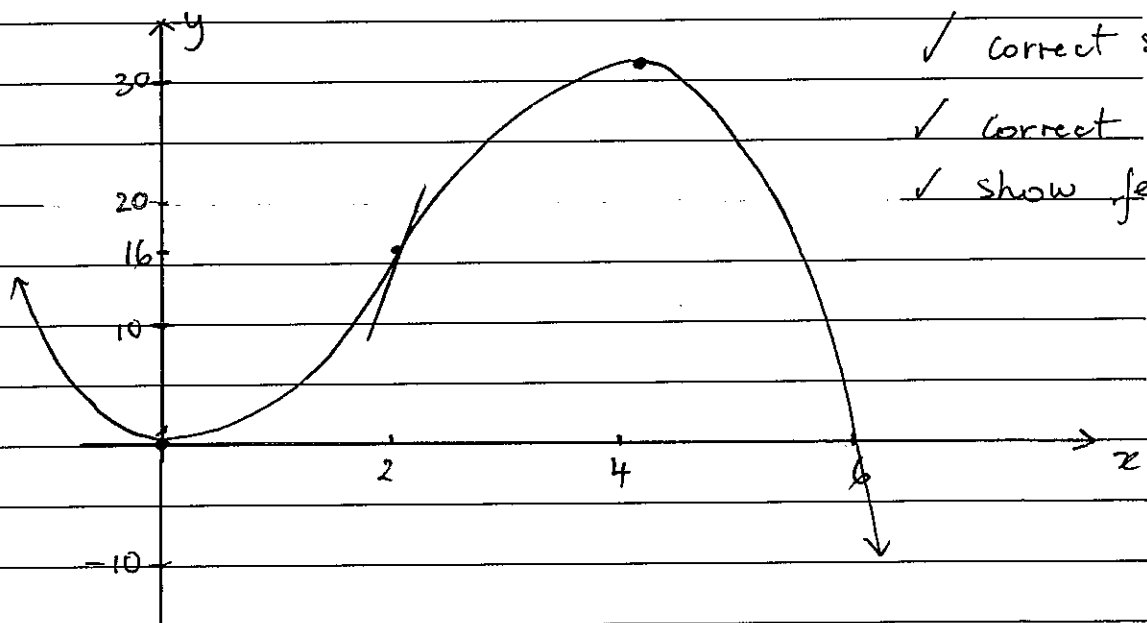
ii) for point of inflexion, $f''(x) = 0$

$$\Rightarrow 12 - 6x = 0$$

$$\begin{cases} x = 2 \\ y = 16 \end{cases}$$

x	1	2	3
$f''(x)$	6	0	-6
	↖		↘

$\therefore$ The point of inflexion is $(2, 16)$



$$\begin{aligned} x\text{-intercept} : 6x^2 - x^3 &= 0 \\ -x^2(x-6) &= 0 \\ x &= 0, \quad x = 6 \end{aligned}$$

iv) The least value of $f(x)$ in the domain

$$-1 \leq x \leq 7 \text{ is}$$

$$y = 6 \times 7^2 - 7^3 = -49 \quad \checkmark$$